SENSOR BASED PLANTS PROTECTION USING ARDUINO

A

Real Time / Field Based Research Project

Submitted in the Partial fulfilment of the

Academic Requirements for the Award of the Degree of

Bachelor of Technology

In

Electronics and Communication Engineering

By

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Under the esteemed guidance of

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ACE

Engineering College

(An UGC Autonomous Institution)

Ankushapur (V), Ghatkesar (M), R.R.Dist – 501 301

(Approved by AICTE, New Delhi and Affiliated to JNTUH)

NBA Accredited B.Tech Courses: EEE, ECE, CSE, CIVIL, MECH

Accredited by NAAC 'A' Grade

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CERTIFICATE

This is to certify that the project work entitled **"SENSOR BASED PLANTS PROTECTION SYS TEM USING ARDUINO"** done by

B. YOKSHITHA	**23AG1A0403**

Department of Electronics and Communication Engineering, is a record of Bonafide work carried out by her. This Project is done as partial fulfilment of obtaining Bachelor of Technology degree to be awarded by **JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY, HYDERABAD** during the academic year 2024-2025.

Mr. P. SUBRAHMANYAM	**Dr. Y. CHAKRAPANI**
Associate Professor	**Professor and DEAN**

ACKNOWLEDGEMENT

I am happy for being guided by **Mr.P.SUBRAHMANYAM, Associate Professor** for his able guidance given to us to complete our Real Time Project / Field Based Project work successfully.

I wish to express our deep sense of gratitude to our **Project coordinator Mrs. CH.SARITA, Assistant Professor** for giving us an opportunity to present the technical Project work.

I am grateful to thank **Dr.Y. CHAKRAPANI**, Professor and Dean, Dept. of ECE, for his thought-provoking suggestions and constant encouragement that led us to this success.

I would like to convey our heartfelt thanks to Principal sir for the wonderful guidance and encouragement given to us to move ahead in the execution of this project and **Prof. Y. V. GOPALA KRISHNA MURTHY sir**, for his moral support and providing infrastructure.

Above all, I am very much thankful to the management of **ACE Engineering College** which was established by the high-profiled intellectuals for the cause of Technical Education in the modern era.

I am thankful to **JNTUH** for giving us this opportunity to work on a project as a part of our curriculum.

I take this opportunity to thank one and all who have helped in making this Real Time Project/ Field Based Project possible.

With Regards:

B. YOKSHITHA 23AG1A0403

ABSTRACT

The Protecting crops from rain-induced damage is critical for ensuring agricultural sustainability, particularly in regions prone to heavy rainfall. This abstract examines various strategies and technologies aimed at mitigating the adverse effects of rain on crop health and yield. Traditional methods such as crop rotation and mulching are complemented by modern advancements including rain-resistant crop varieties and precision agriculture techniques Integrated approaches combining these methods effectively minimize soil erosion and waterlogging, and moisture stress, thereby safeguarding agricultural productivity contributing to food security in rain-prone areas. This real-time crop protection project aims to safeguard crops from the adverse effects of rainfall. By integrating weather prediction technology with automated protective measures, the system can monitor and respond to weather conditions in real time. Utilizing sensors, weather forecasting data, it predicts rainfall and activates protective mechanisms like retractable covers and drainage systems. The system includes a weather prediction module, sensor network, control system, actuation mechanisms, and a communication network. Field trials will measure its effectiveness in enhancing crop health and yield. This innovative solution provides farmers with a proactive tool to mitigate rain damage, promoting sustainable agriculture and economic stability.

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CHAPTER 1

INTRODUCTION

1.1 Introduction of the project

Farming is the science and art of creating plants and creatures. Agriculture was the key headway in the climb of stationary human advancement whereby developing of prepared species made sustenance surpluses that engaged people to live in urban regions. The chronicled background of agriculture began an enormous number of years earlier. Consequent to get- together wild grains beginning at any rate 1,05,000 years earlier, early farmers began to plant them around 11,500 years back. Pigs, sheep, and cows were prepared over 10,000 years earlier. Plants were autonomously developed in any event 11 districts of the world. Modern horticulture dependent for enormous scope mono culture in the twentieth century came to overwhelm agrarian yield, however around two billion individuals despite the fact that everything relied up on subsistence agribusiness into the twentyfirst..

The natural impact of agribusiness is the effect that diverse cultivating practices have on the general conditions, and how those effects can be followed back to those practices. The characteristic impact of agribusiness fluctuates reliant on the wide combination of agricultural practices used over the world. Finally, the normal impact depends upon the creation practices of the system used by farmers. The relationship between spreads into the earth and the developing system is convoluted, in this manner it depends upon other air factors such as precipitation and temperature.

There are two kinds of markers of natural impact: "signifies based", which relies upon the farmer's creation procedures, and "impact based", which is the impact that developing strategies have on the cultivating structure or spreads to the earth. An instance of a strategies-based marker would be the idea of groundwater that is impacted by the proportion of nitrogen applied to the earth. A marker reflecting the loss of nitrate to groundwater would be impact based. The techniques-based on assessment realises farmers demonstrations of agribusiness, and the impact-based assessment deliberates the genuine effects of the agrarian system. For example, the methods -based analysis may look at pesticides and treatment systems that farmers are using, and impact-based assessment would consider the CO2 which is being released or what the Nitrogen substance of the soil is.

The natural effect of agribusiness includes an assortment of components from the dirt, to water, the air. creature and soil assortment, individuals, plants and the nourishment itself. A portion of the ecological issues that are identified with farming are environmental change, deforestation, no man's lands, hereditary building, water system issues, contamination, soil corruption, and waste.

These days, during the stormy season the developed harvests get influenced because of overwhelming precipitation. The proposed framework includes security of the harvests via auto rooftop which covers the specific region.

Hence in order to close the rooftop, the farmer needs to send an SMS to GSM module, When downpour is halted, controller consequently opens the rooftop This research paper is more acquainted with objective and motivation behind the proposed method. Collected fields influenced or demolished because of substantial downpour and shortage issue. Various existing systems are discussed in the literature survey also listed its practical views and explained proposed strategy to defeat the impediments of the current one and giving best outcomes to farmers. The significant objective of this paper is to keep the reaped crops from the overwhelming a precipitation and spare the downpour water. The rain sensor is utilized for the working of rooftop when there is precipitation.

1.2 LITERATURE SURVEY

We've examined work on similar projects in the past and know how to do research. Agriculture is the back bone of the Indian economy, according to many ways. Agriculture is the primary source of nourishment for us, making life impossible without it. However, in the current situation, finding farm employees is difficult. Modern development is prompted by the computerization of all industries. Up to a certain extent, the agricultural process is automated here.

P. Goutham Goud et al [1] Rain sensor, a sophisticated microprocessor, and a DC motor are used in a system where the deluge is recognised and a protective shield is wrapped around the rooftop. The rain sensor of such a drying shed protects the harvest from rain and wetness. To automate this task, a rainfall detects the downpour and sends the information to the microcontroller. A defensive wrapper is wrapped over the rooftop top, and the microcontroller forms the information and activates the DC motor control circuit.

Dheekshith et al [2] developed a system for identifying precipitation by using a downpour distinguishing sensor. The sensor is connected to a direct actuator motor and a spread job that protects against rain. When the sensor detects rain, it goes to work and pivots the spreading roll, which covers the gathered merchandise and protects the farmer from losses.

Naveen K B et al [3] suggested a framework that was structured using the Proteus programming language. When the rain sensor detects a deluge, the soil moisture sensor determines dampness content, which is displayed on the LCD. . The value sent to the PIC microcontroller is determined by the soil moisture sensor, temperature sensor, and rain sensor. The automatic rainwater and crop saving system protect crops from excessive rainwater by taking into consideration the attributes. The current work entails preserving the unique resources that are available to mankind. We can limit the flow of water and so eliminate waste by assessing the status of the soil productively. Water stream can be obligated by substantially sending by knowing the state of moistness, and temperature over with the use of unexpectedness and temperature sensors. There are currently no effective frameworks available in the current situation. The farmer must go to the drying region and cover the gathered fields, which is particularly difficult if the farmer's location is distant from the harvest and the entire crop would be pummelled by the downpour before the farmer arrives.

Limitation of Existing Systems:

The greenhouse has certain limitations such as it cannot be adjusted to climatic conditions and it will not allow the sun to pass through the crop as shown in figure 3. Most of the existing systems concentrated on covering the crop from rainfall by providing a roof over the crop rather than focusing on the water level required for a particular crop. If the crop needs a certain amount of water in such case farmer needs to permit the rain over the crop and once it’s having sufficient amount of water content then he can protect the crop by covering it with a roof. Though we preserve the rainwater during rainfall, later if the farmer wants to use it for the crops it consumes more electrical power.

1.2 Proposed System:

The proposed system for crop protection from rain integrates weather prediction, real-time monitoring, and automated response mechanisms. It employs a network of sensors to collect data on soil moisture, rainfall, and temperature. This data, combined with advanced weather forecasting, allows the system to predict imminent rainfall. Upon detecting potential harmful rain, the control system triggers protective measures such as retractable covers and drainage systems to shield crops. The system ensures seamless communication between sensors, control units, and actuators, enabling timely and efficient responses. Field trials will validate its effectiveness in enhancing crop resilience and yield, providing a scalable solution for various agricultural settings.

CHAPTER 2

HARDWARE AND SOFTWARE DESCRIPTION

2.2.1 ARDUINO OVERVIEW

Arduino is a prototype platform based on an easy-to-use hardware and software. It consists of a circuit board, which can be programmed and a ready-made software called Arduino IDE (Integrated Development Environment), which is used to write and upload the computer code to the physical board.

The key features are:

- Arduino boards are able to read analog or digital input signals from different sensors and turn it into an output such as activating a motor, turning LED on/off, connect to the cloud and many other actions.
- You can control your board functions by sending a set of instructions to the microcontroller on the board via Arduino IDE (referred to as uploading software).
- Unlike most previous programmable circuit boards, Arduino does not need an extra piece of hardware in order to load a new code onto the board. You can simply use a USB cable.
- Additionally, the Arduino IDE uses a simplified version of C++, making it easier to learn to program.

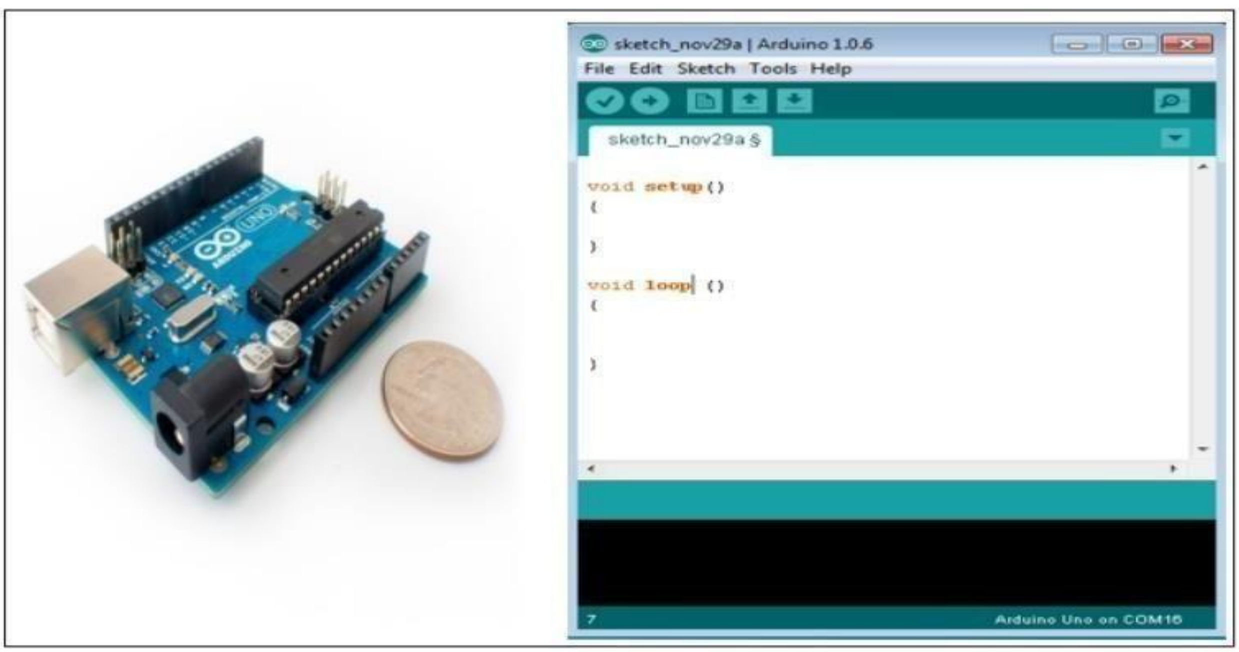

Fig 2.1 Arduino Uno

2.1.2 Arduino Uno Description

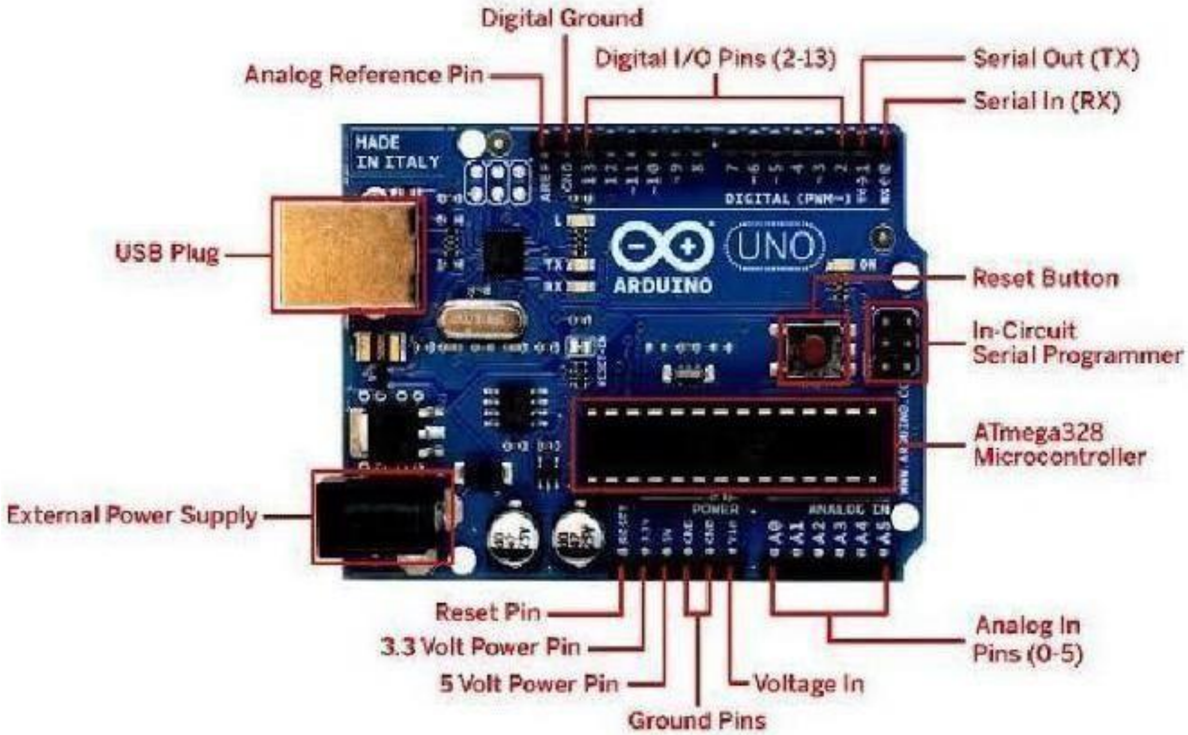

Fig2.2: Arduino Uno Description

- **Power USB:**

Arduino board can be powered by using the USB cable from your computer. All you need to do is connect the USB cable to the USB connection (1).

- **Power (Barrel Jack):**

Arduino boards can be powered directly from the AC mains power supply by connecting it to the Barrel Jack (2).

- **Voltage Regulator:**

The function of the voltage regulator is to control the voltage given to the Arduino board and stabilize the DC voltages used by the processor and other elements.

- **Crystal Oscillator:**

The crystal oscillator helps Arduino in dealing with time issues. How does Arduino calculate time? The answer is, by using the crystal oscillator. The number printed on top of the Arduino crystal is 16.000H9H. It tells us that the frequency is 16,000,000 Hertz or 16MHz**.**

- **Arduino Reset:**

You can reset your Arduino board, i.e., start your program from the beginning. You can reset the UNO board in two ways. First, by using the reset button (17) on the board. Second, you can connect an external reset button to the Arduino pin labelled RESET (5).

6,7,8,9 Pins (3.3, 5, GND, Vin):

- 3.3V (6): Supply 3.3 output volt
- 5V (7): Supply 5 output volt
- Most of the components used with Arduino board works fine with 3.3 volt and 5 volt.
- GND (8)(Ground): There are several GND pins on the Arduino, any of which can be used to ground your circuit.
- Vin (9): This pin also can be used to power the Arduino board from an external power source, like AC mains power supply**.**

- **Analog pins:**

The Arduino UNO board has five analog input pins A0 through A5. These pins can read the signal from an analog sensor like the humidity sensor or temperature sensor and convert it into a digital value that can be read by the microprocessor.

- **Main microcontroller:**

Each Arduino board has its own microcontroller (11). You can assume it as the brain of your board. The main IC (integrated circuit) on the Arduino is slightly different from board to board. The microcontrollers are usually of the ATMEL Company. You must know what IC your board has before loading up a new program from the Arduino IDE. This information is available on the top of the IC. For more details about the IC construction and functions, you can refer to the data sheet.

- **ICSP pin:**

Mostly, ICSP (12) is an AVR, a tiny programming header for the Arduino consisting of MOSI, MISO, SCK, RESET, VCC, and GND. It is often referred to as an SPI (Serial Peripheral Interface), which could be considered as an "expansion" of the output. Actually, you are slaving the output device to the master of the SPI bus. **Power LED indicator:**

This LED should light up when you plug your Arduino into a power source to indicate that your board is powered up correctly. If this light does not turn on, then there is something wrong with the connection.

- **TX and RX LEDs:**

On your board, you will find two labels: TX (transmit) and RX (receive). They appear in two places on the Arduino UNO board. First, at the digital pins 0 and 1, to indicate the pins responsible for serial communication. Second, the TX and RX led (13). The TX led flashes with different speed while sending the serial data. The speed of flashing depends on the baud rate used by the board. RX flashes during the receiving process.

- **Digital I / O:**

The Arduino UNO board has 14 digital I/O pins (15) (of which 6 provide PWM (Pulse Width Modulation) output. These pins can be configured to work as input digital pins to read logic 14 values (0 or 1) or as digital output pins to drive different modules like LEDs, relays, etc. The pins labelled “~” can be used to generate PWM.

- **AREF:**

AREF stands for Analog Reference. It is sometimes, used to set an external reference voltage (between 0 and 5 Volts) as the upper limit for the analog input pins.

2.1.1 Hardware Specifications:

1. Microcontroller: ATmega328P

2. Operating Voltage: 5V

3. Input Voltage (recommended): 7-12V

4. Input Voltage (limit):6-20V

5. Digital I/O Pins:14 (of which 6 can be used as PWM outputs)

6. Analog Input Pins: 6

7. DC Current per I/O Pin: 20 mA

8. DC Current for 3.3V Pin:50 mA

9. Flash Memory:32 KB (ATmega328P) of which 0.5 KB is used by the bootloader

10. SRAM: 2 KB (ATmega328P)

11. EEPROM: 1 KB (ATmega328P)

12. Clock Speed:16 MHz

13. LED_BUILTIN: Pin 1

14. Top of Form

MICROCONTROLLER

The microcontroller used on the Arduino board is essentially used for controlling all major operations. The microcontroller is used to coordinate the input taken and execute the code written in a high-level language. This code is then implemented and relevant output is generated. The choice of microcontroller varies on the requirements of the project.

The microcontroller used in the Arduino shown above is ATmega328P manufactured by ATMEL Company and it is the most common choice. Here are some features of this microcontroller.

- ATmega328P has 14 Digital I/O Pins. Out of the 14 pins, 6 provide PWM(pulse width modulation) output.

- ATmega328P can have 6 (DIP) or 8 (SMD) Analog Input Pins.

- The DC Current which is supplied to each I/O Pin is around 40 mA.

- ATmega328P has a flash memory of 32 KB.

- ATmega328P has a SRAM (Static Random-Access Memory) of 2 KB.

- ATmega328P has EEPROM (Electrically Erasable Programmable Read-Only Memory) of 1 KB.

2.1.2 Connectivity

The board has a USB port for communication with a computer, which also supplies power. Additionally, it features a power jack for external power supplies, and a reset button. The USB interface uses the ATmega16U2 (or ATmega8U2 in older revisions) for serial communication. This interface allows the board to appear as a virtual COM port on a connected computer, enabling easy programming and debugging.

2.1.3 Pins and headers

-**Digital Pins:**14 digital I/O pins can be used for various functions such as reading sensors, controlling LEDs, motors, and other devices. Six of these pins (3, 5, 6, 9, 10, and 11) provide Pulse Width Modulation (PWM) output.
-**Analog Inputs:** 6 analog inputs (A0-A5) that can read signals from analog sensors and convert them into digital values.
- **ICSP Header:** The In-Circuit Serial Programming (ICSP) header allows for programming the microcontroller without the need for a bootloader.
- **Reset Button-** A physical button to reset the board.

2.1.5 Shields and modules

One of the key features of the Arduino ecosystem is the availability of shields—modular boards that can be plugged on top of the Arduino board to extend its functionality. Shields can add capabilities such as wireless communication, motor control, GPS, and more. Additionally, a wide range of sensors and modules can be connected to the Arduino Uno to interact with the physical world.

2.1.6 Community and Resources

Arduino has a large, active community of users and developers. This community provides extensive documentation, tutorials, forums, and projects that cater to a wide range of interests and skill levels. The open-source nature of the platform encourages collaboration and innovation, making it an excellent resource for learning and experimentation.

For building a sensor-based plant protection system with Arduino, there are several community resources and active platforms you can explore:

1. Arduino Forum: This is a great place to ask questions, share your projects, and get advice from experienced Arduino users. You can find discussions related to sensors, modules, and plant care projects.

2. Instructables: This website has a wealth of step-by-step guides and tutorials on various Arduino projects, including plant monitoring systems. You can find projects that utilize the sensors and modules mentioned earlier.

3. Hackster.io: This is another platform where makers share their projects. You can search for plant monitoring or protection systems to find inspiration and detailed instructions.

4. YouTube: There are countless tutorials on YouTube that cover everything from setting up specific sensors to creating complete plant care systems. Channels focused on Arduino projects often have valuable content.

5. Reddit: Subreddits like r/Arduino and r/DIY can be helpful for connecting with other hobbyists. You can ask for tips, share your progress, and get feedback on your project.

6. GitHub: Many developers share their code and projects related to Arduino on GitHub. You can find libraries and example projects that can help you with your plant protection system.

7. Arduino Project Hub: This is an official Arduino resource where users share their projects. You can find various projects related to gardening and plant care, complete with parts lists and code.

These resources should provide you with a solid foundation and community support as you work on your project! If you have any specific questions or need further assistance, just let me know.

2.1.4 SOFTWARE AND HARDWARE SPECIFICATION DOCUMENT

1. Microcontroller - Arduino Uno and Arduino Ide Software

The Arduino Interfaced Development Environment, or more also known as the Arduino Ide software (IDE), is indeed available. It includes a scripting language for a scripting language, a user interface, a text terminal, a toolbox with buttons for frequently performed functions, and a variety of menus. It interfaces with the Arduino hardware to upload applications and communicate with them.

Fig 2.3: **Arduino IDE and Arduino UNO Board**

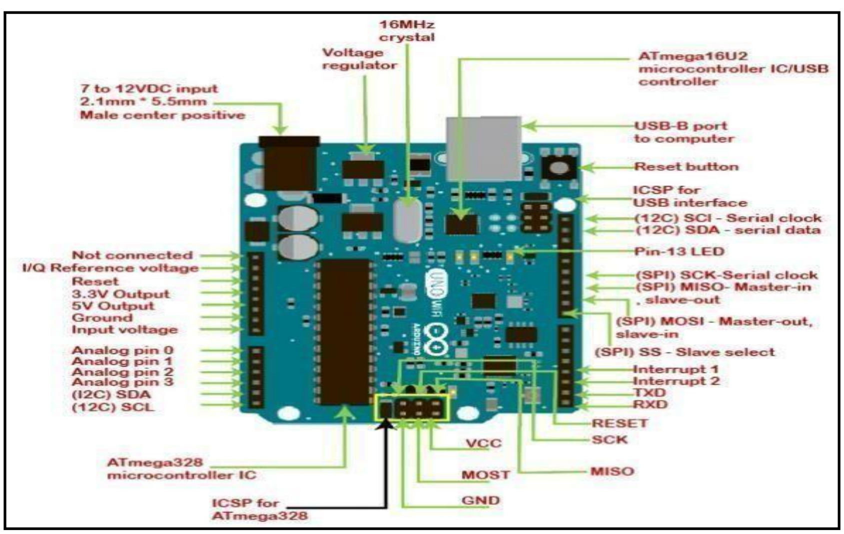

Fig 2.4: Pin Description Arduino UNO Board

2.1.5 ARDUINO – INSTALLATION

After learning about the main parts of the Arduino UNO board, we are ready to learn how to set up the Arduino IDE. Once we learn this, we will be ready to upload our program on the Arduino board. In this section, we will learn in easy steps, how to set up the Arduino IDE on our computer and prepare the board to receive the program via USB cable.

Step 1: First you must have your Arduino board (you can choose your favourite board) and a USB cable. In case you use Arduino UNO, Arduino Duemilanove, Nano, Arduino Mega 2560, or Diecimila, you will need a standard USB cable (A plug to B plug), the kind you would connect to a USB printer as shown in the following image.

Fig 2.5: USB cable

In case you use Arduino Nano, you will need an A to Mini-B cable instead as shown in the following image.

Step 2: Download Arduino IDE Software:

You can get different versions of Arduino IDE from the Download page on the Arduino Official website. You must select your software, which is compatible with your operating system (Windows, IOS, or Linux). After your file download is complete, unzip the file.

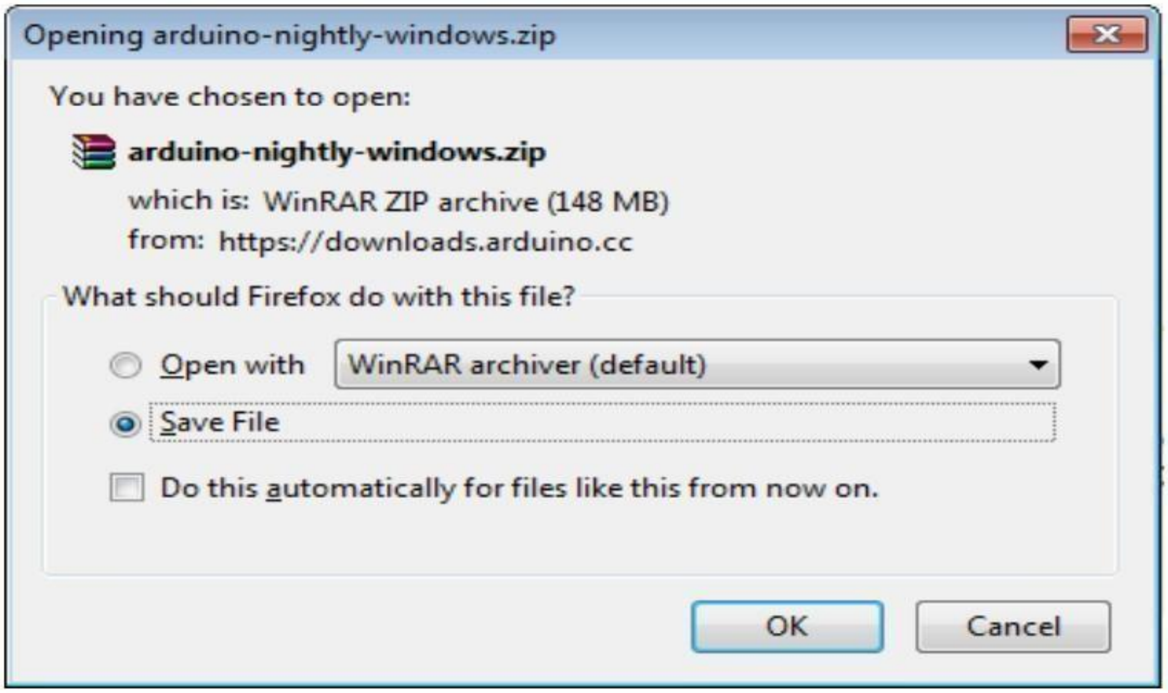

Figure 2.5(a) Download Arduino IDE Software

Step 3: Power up your board:

The Arduino Uno, Mega, Duemilanove and Arduino Nano automatically draw power from either, the USB connection to the computer or an external power supply. If you are using an Arduino Diecimila, you have to make sure that the board is configured to draw power from the USB connection. The power source is selected with a jumper, a small piece of plastic that fits onto two of the three pins between the USB and power jacks. Check that it is on the two pins closest to the USB port. Connect the Arduino board to your computer using the USB cable. The green power LED (labelled PWR) should glow.

Step 4: Launch Arduino IDE:

After your Arduino IDE software is downloaded, you need to unzip the folder. Inside the folder, you can find the application icon with an infinity label (application.exe). Double-click the icon to start the IDE.

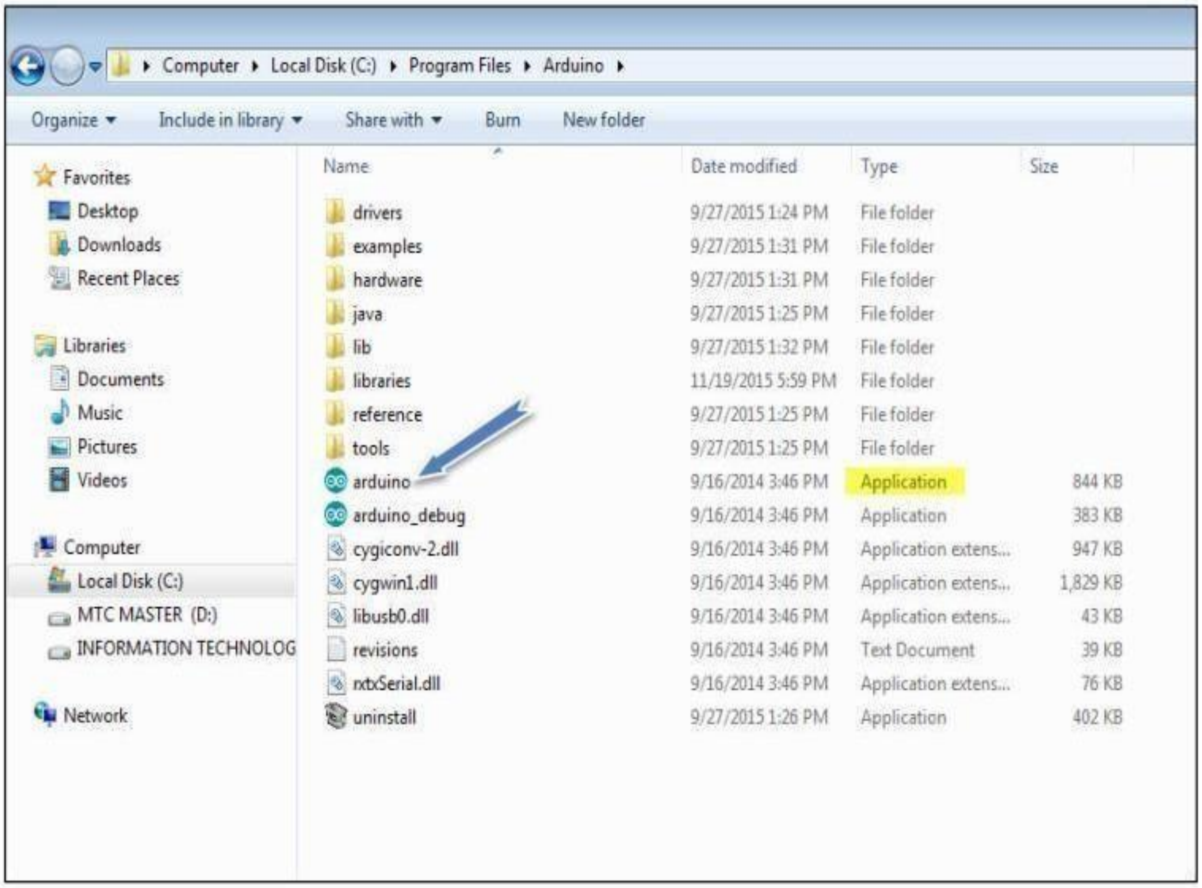

Figure 2.5(b) Launch Arduino IDE

Step 5: Open your first project:

Once the software starts, you have two options:

- Create a new project.
- Open an existing project example. To create a new project, select File --> New.

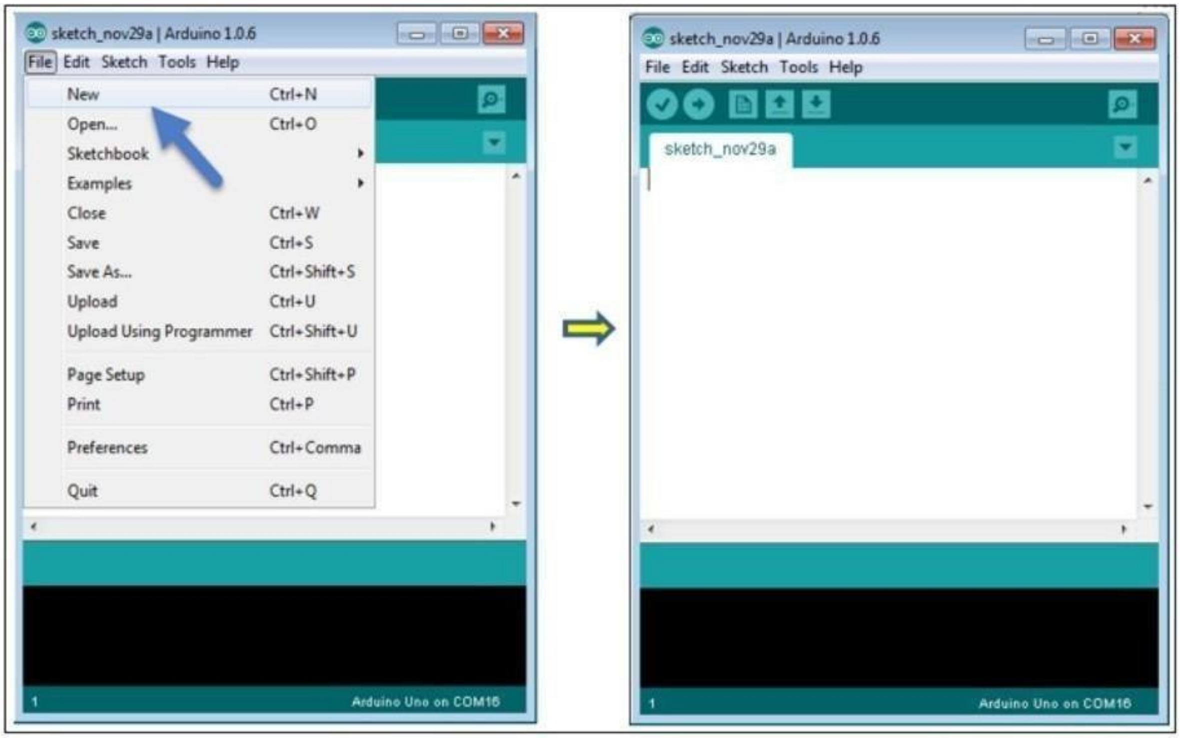

Figure 2.5(c) To open New Project

To open an existing project example, select File -> Example -> Basics -> Blink.

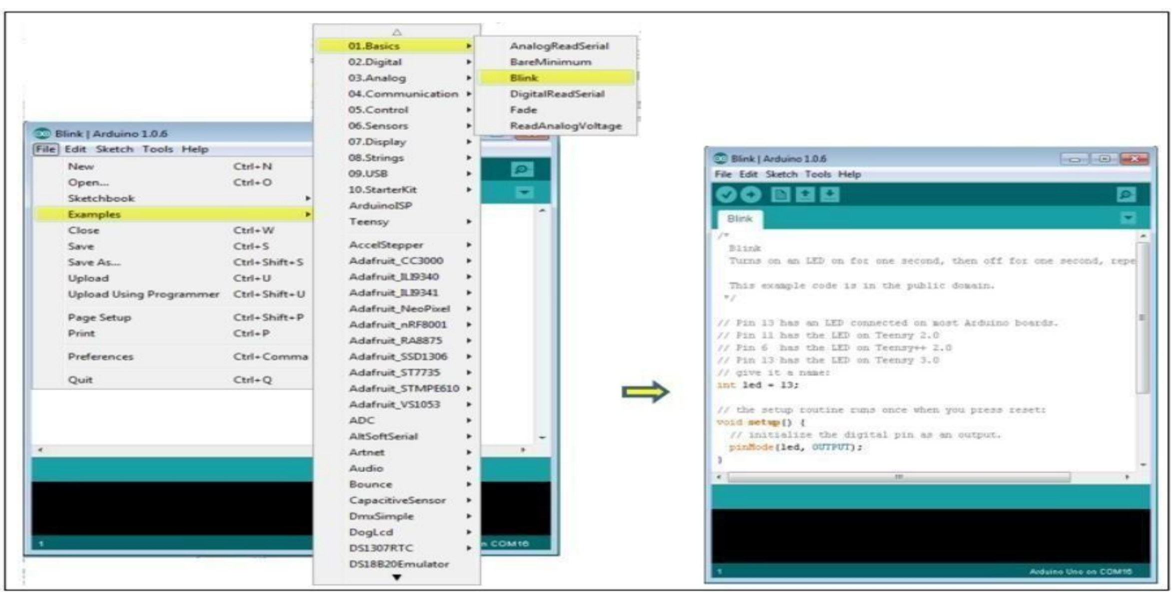

Figure 2.5(d) To open an existing project example

Here, we are selecting just one of the examples with the name Blink. It turns the LED on and off with some time delay. You can select any other example from the list.

Step 6: Select your Arduino board:

To avoid any error while uploading your program to the board, you must select the correct Arduino board name, which matches with the board connected to your computer. Go to Tools -> Board and select your board.

2.2.4 PROGRAM

```
#include <Servo.h>
int rain_sensor = 7;
int light_sensor = 6;
int servo1 = 9;
int servo2 = 11;
Servo myServo1;
Servo myServo2;
void setup() {
// Set up sensor pins asinput
pinMode(rain_sensor,INPUT);
pinMode(light_sensor, INPUT);
Serial.begin(9600);
// Attach servos to pins and set initial position
myServo1.attach(servo1);
myServo2.attach(servo2);
myServo1.write(0); myServo2.write(0);
}
void loop() { // Read sensor values
int sensorvalue1 = digitalRead(rain_sensor);
Serial.println("rain sensor");
Serial.println(sensorvalue1);
int sensorvalue2 = digitalRead(light_sensor);
Serial.println("light sensor");
Serial.println(sensorvalue2);
if ((sensorvalue1 == 0)&&(sensorvalue2 == 0)) {
// Rotate both servos to 180 degrees if rain sensor is triggered
myServo1.write(180);
myServo2.write(180);
```

```
Serial.println("Servos are on");
delay(500);
}
else if ((sensorvalue1 == 0)&&(sensorvalue2 == 1)) {
// Rotate both servos to 180 degrees if rain sensor is triggered
myServo1.write(180);
myServo2.write(180);
Serial.println("Servos are on");
delay(500);
}
else if ((sensorvalue1 == 1)&&(sensorvalue2 == 1)) {
// Rotate both servos to 180 degrees if rain sensor is triggered
myServo1.write(180);
myServo2.write(180);
Serial.println("Servos are on");
delay(500);
}
else  {
// Reset servos to 0 degrees if light sensor is triggered
myServo1.write(0);
myServo2.write(0);
Serial.println("Servos are off");
delay(500);
}

}
```

CHAPTER 3
PROJECT DESCRIPTION

3.1 Block Diagram

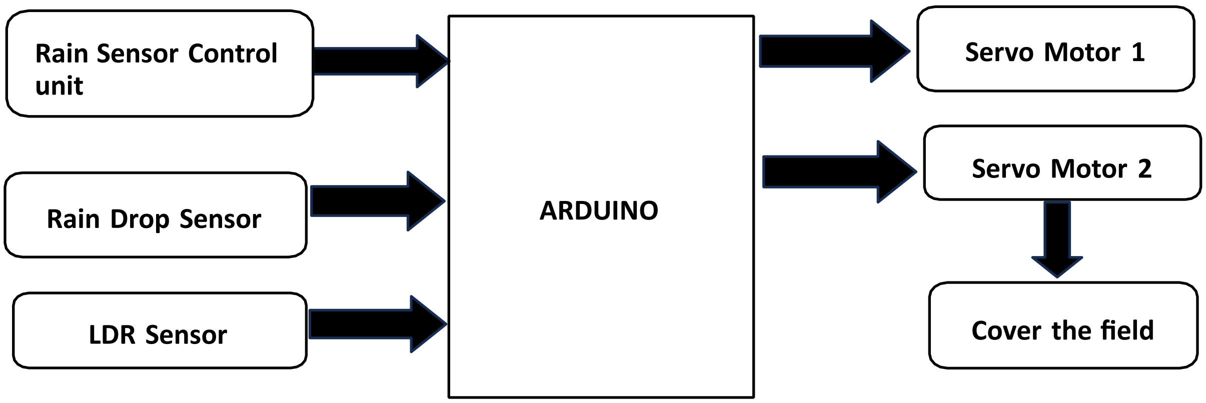

Fig 3.1: Basic Block Diagram for Crop protection from rain

3.2 Circuit Diagram

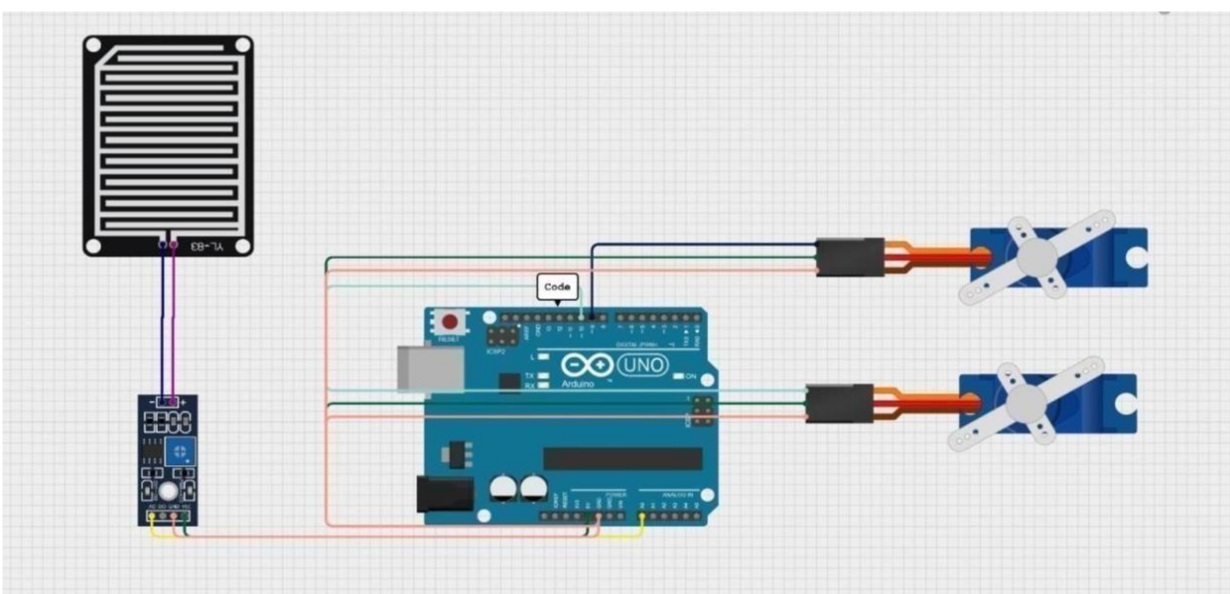

Fig 3.2: Circuit Diagram of Crop protection from rain

3.2.1 Rain drop Control Unit

A raindrop sensor operates based on the principle of conductivity. Typically constructed with exposed conductive traces or pads on a surface, the sensor detects the presence and intensity of raindrops by measuring changes in conductivity between these elements. When rain falls on the sensor surface, water droplets act as conductive bridges between the traces or pads, altering the electrical resistance or capacitance of the circuit. This change is then converted into an electrical signal, often analog, proportional to the amount of water detected. Some sensors also provide digital outputs that indicate the presence or absence of rain above a certain threshold. This real-time data is valuable for various applications such as weather monitoring systems, automated irrigation systems, and smart home automation, enabling timely responses to weather conditions based on accurate rainfall detection.

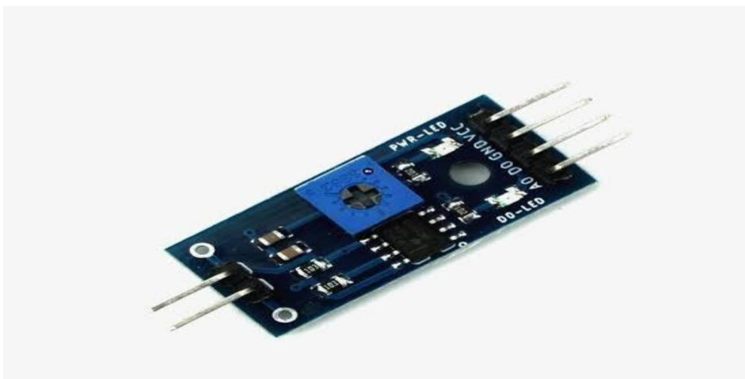

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Fig 2.3: Rain drop Control Unit

3.2.2 Servo motors

A servo motor works by receiving a control signal that represents a desired output position. This signal is typically a Pulse Width Modulation (PWM) signal. Inside the servo, there is a small circuit that compares the received signal to the position of a potentiometer linked to the motor shaft. If there is a discrepancy between the desired and actual positions, the circuit adjusts the motor's position by rotating the shaft in the appropriate direction. This adjustment continues until the actual position matches the desired position. The system uses feedback to ensure accurate control of linear position, velocity, and acceleration. Servo motors are commonly used in applications requiring precise control, such as robotics and automation.

Servo Motor working Mechanism

It consists of three parts:

1. Controlled device
2. Output sensor
3. Feedback system

It is a closed-loop system where it uses a positive feedback system to control motion and the final position of the shaft. Here the device is controlled by a feedback signal generated by comparing output signal and reference input signal.

Here reference input signal is compared to the reference output signal and the third signal is produced by the feedback system. And this third signal acts as an input signal to the control the device. This signal is present as long as the feedback signal is generated or there is a difference between the reference input signal and reference output signal. So the main task of servomechanism is to maintain the output of a system at the desired value at presence of noises.

Servo Motor Working Principle

A servo consists of a Motor (DC or AC), a potentiometer, gear assembly, and a controlling circuit. First of all, we use gear assembly to reduce RPM and to increase torque of the motor. Say at initial position of servo motor shaft, the position of the potentiometer knob is such that there is no electrical signal generated at the output port of the potentiometer. Now an electrical signal is given to another input terminal of the error detector amplifier. Now the difference between these two signals, one comes from the potentiometer and another comes from other sources, will be processed in a feedback mechanism and output will be provided in terms of error signal. This error signal acts as the input for motor and motor starts rotating. Now motor shaft is connected with the potentiometer and as the motor rotates so the potentiometer and it will generate a signal. So as the potentiometer's angular position changes, its output feedback signal changes. After sometime the position of potentiometer reaches at a position that the output of potentiometer is same as external signal provided. At this condition, there will be no output signal from the amplifier to the motor input as there is no difference between external applied signal and the signal generated at potentiometer, and in this situation motor stops rotating.

Fig 2.4 : Servo motors

3.2.3 Rain Drop Sensor

The raindrop sensor module is used for rain detection. It is also for measuring rainfall intensity. The module includes a rain board and a control board that are separate for more convenience. It has a power indicator LED and an adjustable sensitivity though a potentiometer. The module is based on the LM393 op amp. It includes a printed circuit board(control board) that "collects" the rain drops. As rain drops are collected on the circuit board, they create paths of parallel resistance that are measured via the op amp. The lower the resistance (or the more water), the lower the voltage output. Conversely, the less water, the greater the output voltage on the analog pin. A completely dry board for example will cause the module to output five volts.

A rain sensor control unit is an electronic device designed to detect the presence of rain and manage connected systems accordingly, such as closing windows, retracting roofs, or pausing irrigation systems

Fig: 2.5 Rain Drop Sensor

Specifications of Raindrop sensor:

- Operating voltage: 5V
- Provide both digital and analog output
- Adjustable sensitivity
- Output LED indicator
- Compatible with Arduino
- TTL Compatible
- Bolt holes for easy installation

3.2.4 LDR

LDR full form Light Dependent Resistor. LDR is a simple device that can be used to detect light levels and respond to light. When there's more light, its resistance goes down, and when there's less light, its resistance goes up. LDR light sensor is also known as a photoresistor, photocell, or photoconductor. [1]

The working principle of an LDR is photoconductivity, which is nothing but an optical phenomenon. When the light is absorbed by the material then the conductivity of the material enhances. When the light falls on the LDR, then the electrons in the valence band of the material are eager to the conduction band.

A light dependent resistor (LDR) sensor is a passive electronic component that detects light and measures its intensity. LDRs are also known as photoresistors.

How it works

- LDRs are made of semiconductor materials with two conductors separated by an insulator.
- When light hits the LDR, the insulator becomes more conductive.
- This changes the resistance of the LDR, which decreases as the light intensity increases.
- The LDR converts changes in light into electrical signals that can be measured.

Applications:

- LDRs are used in many applications, including:

o **Light meters**: Measure the amount and brightness of light o

Burglar alarm systems: Detect changes in light to trigger alarms o

Street lighting systems: Automatically turn on lights at night and off at

dawn • LDRs are cost-effective, simple, and easy to integrate into

electronic devices.

Fig : LDR Sensor

3.2.5 Working Principal

To create a crop protection system using Arduino Uno, a rain drop sensor, a rain sensor control unit, two servo motors, and jumper wires, you can follow these steps. This system will help protect crops by covering them with a tarp or similar material when rain is detected.
This code snippet appears to control a simple crop protection system that uses a rain sensor and two servo motors. Here's an explanation of how it works:

1. Components and Setup:

- Rain Sensor: The rain sensor is connected to analog pin A0. It detects the presence and intensity of rain.
- Servo Motors: Two servo motors are connected to digital pins 9 and 10. These motors likely control some physical barrier or cover that protects the crops.

2. Initialization (`setup` function):

- The servos are attached to their respective pins.
- The initial position of both servos is set to 0 degrees, which might represent the open position of the protective cover.

3. Main Loop (`loop` function):

- The code continuously reads the analog value from the rain sensor. - The `map` function scales the sensor reading (which ranges from 220 to 1023) to a servo angle range of 180 to 0 degrees. This means:
- When the sensor reads 220 (indicating no rain or very light rain), the servos will be at 180 degrees (potentially fully open or retracted).
- When the sensor reads 1023 (indicating heavy rain), the servos will be at 0 degrees (potentially fully closed or extended).

The calculated angle is then sent to both servos, adjusting their positions accordingly. - A delay of 1000 milliseconds (1 second) is included to slow down the loop, giving the servos time to move and preventing the system from reacting too quickly.

CHAPTER 4
RESULTS

4.1 Different Cases of Result

Case1: Before Applying water droplets to rain drop sensor without input

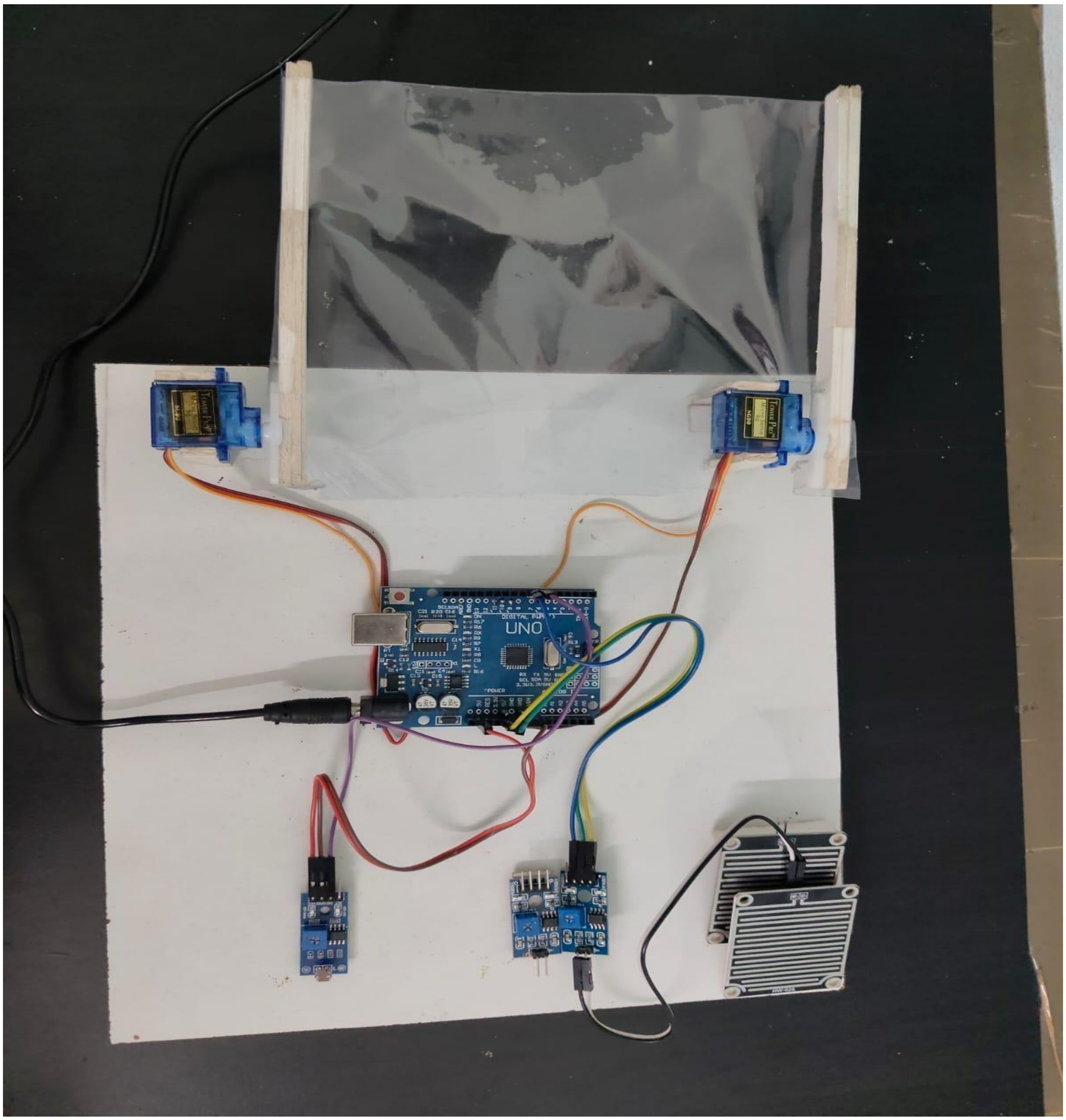

Fig 4.1 Before the water droplet without input

Case2: Before applying water droplets to rain drop sensor with input

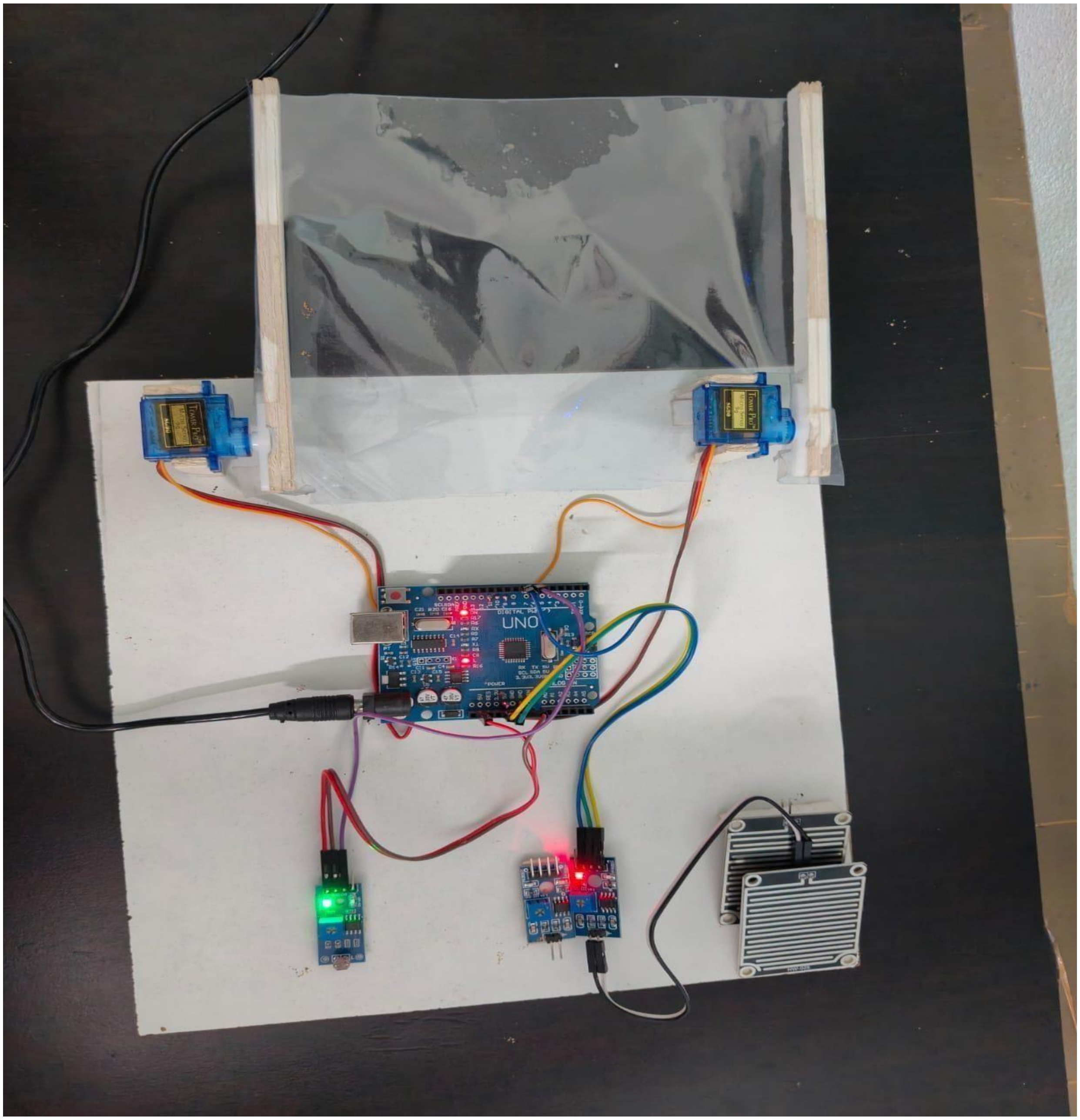

Fig 4.2 Before water droplet with input

Case 3 : After applying water droplet with input

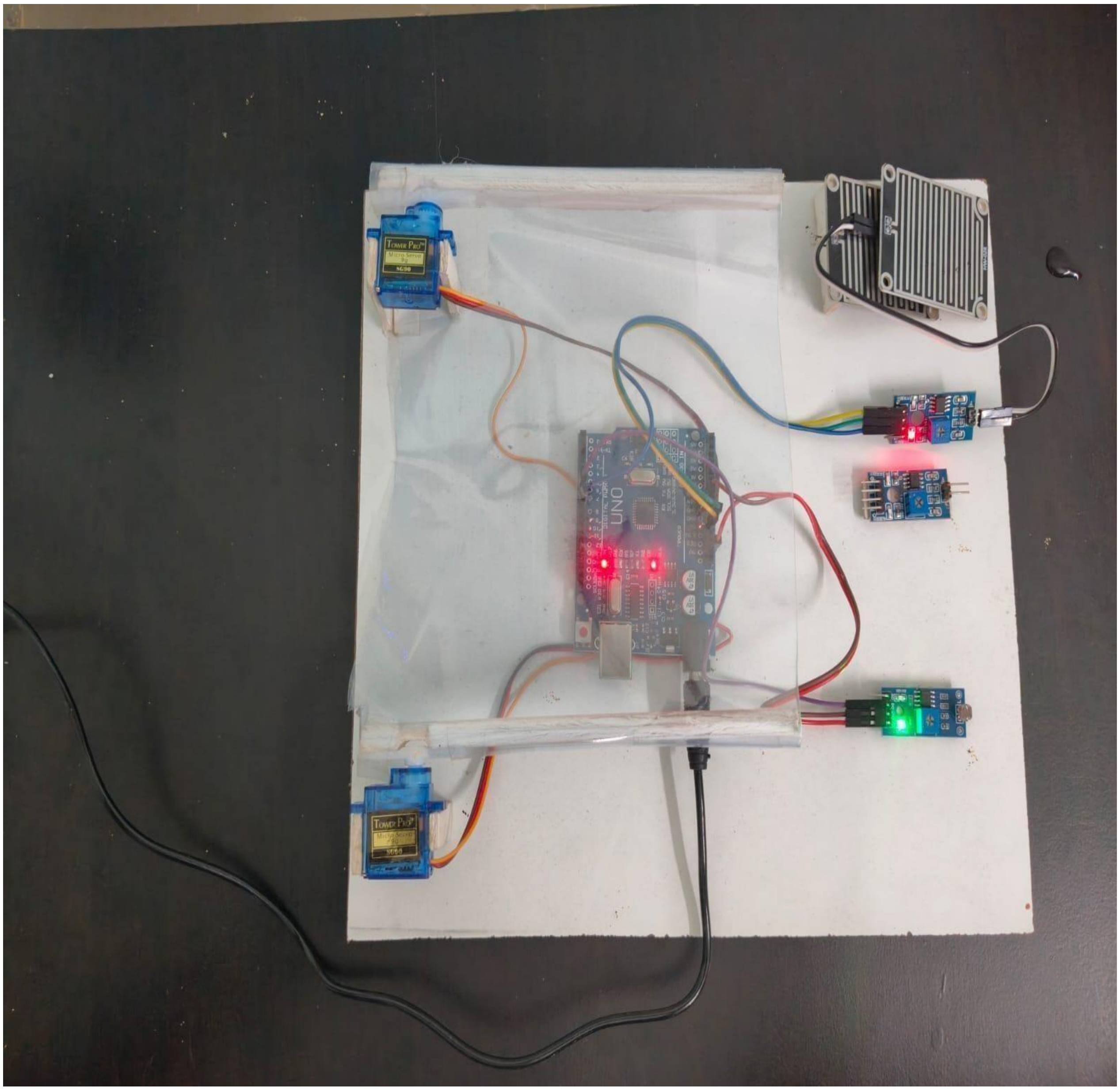

Fig 4.3: after applying water droplet with input

CHAPTER 5
ADVANTAGES, DISADVANTAGES, AND APPLICATIONS

5.1 Advantages

The advantages of sensor-based plant protection using Arduino are numerous and impactful. Here are some key benefits:

1. Real-Time Monitoring: The system continuously collects data from various sensors, allowing for real-time monitoring of environmental conditions. This means any changes can be detected immediately, enabling timely interventions.

2. Automation: By automating processes like watering and temperature control, the system reduces the need for manual labour. This not only saves time but also ensures that plants receive care even when the gardener is not present.

3. Customization: Arduino allows for high levels of customization. Users can tailor the system to meet the specific needs of different plants or crops, adjusting parameters based on individual requirements.

4. Remote Access: Many Arduino-based systems can be connected to the internet, allowing users to monitor and control their plants from anywhere. This feature enhances convenience, especially for those managing large gardens or farms.

5. Resource Efficiency: By optimizing water usage and energy consumption, the system promotes sustainable practices. This means less waste and a lower environmental impact, which is crucial in modern agriculture.

6. Improved Plant Health: By maintaining ideal growing conditions, the system helps enhance plant health, leading to better growth rates and higher yields. This can be particularly beneficial for farmers looking to maximize their output.

7. Cost-Effective: Implementing an Arduino-based system can be more affordable than traditional methods of plant care. The initial investment in sensors and controllers can lead to long-term savings in labor and resources.

8. Data Collection and Analysis: The system can store historical data, allowing users to analyse trends over time. This information can be invaluable for making informed decisions about plant care and improving future practices.

Overall, sensor-based plant protection using Arduino not only enhances the efficiency of plant care but also promotes sustainable agricultural practices, making it a valuable tool for both hobbyists and professional farmers.

5.2 Disadvantages

While sensor-based plant protection using Arduino offers many advantages, there are also some disadvantages to consider:

1. Initial Setup Cost: Although Arduino systems can be cost-effective in the long run, the initial investment for sensors, microcontrollers, and other components can be significant, especially for larger setups.

2. Technical Knowledge Required: Implementing and maintaining an Arduino-based system requires a certain level of technical knowledge. Users need to be familiar with programming and electronics, which may be a barrier for some.

3. System Reliability: Like any technology, sensor-based systems can experience failures or malfunctions. Sensors may become inaccurate over time, or the microcontroller could face issues, leading to potential risks for plant health if not monitored closely.

4. Limited Sensor Range: Depending on the type of sensors used, there may be limitations in terms of range and accuracy. For example, moisture sensors may not provide precise readings if the soil composition varies significantly.

5. Power Dependency: These systems typically rely on a stable power supply. If there are power outages or issues with the power source, the system may fail to operate, potentially putting plants at risk.

6. Complexity of Integration: Integrating multiple sensors and components can become complex, especially if the system needs to communicate with other devices or platforms. This complexity may lead to challenges in troubleshooting and maintenance.

7. Data Overload: While collecting data is beneficial, too much information can be overwhelming. Users may struggle to interpret the data effectively, which could lead to poor decision-making if not managed properly.8. Environmental Limitations: Some sensors may not function optimally under certain environmental conditions, such as extreme temperatures or humidity levels, which could affect the overall effectiveness of the system.

5.3 Applications

Sensor-based plant protection using Arduino has various applications that can significantly enhance agricultural practices. Here are some key applications:

1. Soil Moisture Monitoring: Sensors can measure the moisture level in the soil, allowing farmers to optimize irrigation schedules. This helps in conserving water and ensuring that plants receive the right amount of moisture.

2. Temperature and Humidity Control: By monitoring temperature and humidity levels, farmers can create optimal growing conditions for their plants. This is especially useful in greenhouses, where environmental control is crucial.

3. Pest Detection: Sensors can be used to detect the presence of pests or diseases in crops. For example, image processing techniques can identify changes in plant appearance, signalling potential infestations.

4. Nutrient Management: By monitoring soil nutrient levels, farmers can make informed decisions about fertilization. This helps in providing the right nutrients to plants, enhancing growth and yield.

5. Automated Irrigation Systems: Combining moisture sensors with Arduino allows for the creation of automated irrigation systems that can water plants only when necessary, reducing water waste.

6. Weather Station Integration: Arduino can be used to build a weather station that collects data on local weather conditions, helping farmers make better decisions regarding planting, harvesting, and other agricultural activities.

7. Remote Monitoring and Control: With the use of IoT (Internet of Things) technology, farmers can remotely monitor their crops and control the irrigation systems through smartphones or computers, providing convenience and flexibility.

8. Data Logging and Analysis: Continuous monitoring of various parameters allows for data logging, which can be analysed to identify trends and make more informed decisions about crop management.

These applications demonstrate how sensor-based plant protection using Arduino can lead to more efficient, sustainable, and productive agricultural practices.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

In conclusion, our real-time crop protection from rain project has demonstrated significant potential in mitigating crop damage and optimizing agricultural yield. By utilizing advanced weather forecasting and automated protective measures, we can ensure timely intervention to shield crops from adverse rainfall. This approach not only enhances productivity but also promotes sustainable farming practices. Continued development and implementation of such technologies will be crucial in addressing the challenges posed by climate variability. Overall, our project underscores the importance of integrating innovative solutions to secure the future of agriculture.

Sensor-based plant protection using Arduino represents a transformative innovation in agriculture and horticulture. This system enhances plant care by automating essential processes such as watering, temperature control, and disease prevention. The integration of multiple sensors ensures accurate real-time monitoring, allowing for quick responses to changing environmental conditions. With its cost-effectiveness and ease of implementation, this technology is accessible to both small-scale gardeners and large-scale farmers. The ability to customize and remotely manage plant protection mechanisms adds further convenience and efficiency, reducing the dependency on manual labour. Moreover, the future potential of integrating IoT, AI, and advanced sensors can further refine the effectiveness of this system. By leveraging technology, sensor-based plant protection can lead to improved crop yields, reduced resource wastage, and a more sustainable approach to agriculture. As the demand for smart farming solutions grows, this system holds immense promise in revolutionizing modern agricultural practices and ensuring food security in the long run. Ultimately, the use of Arduino in plant protection marks a significant step toward precision agriculture, paving the way for a smarter, more efficient, and environmentally friendly farming future.

6.2 Future Scope:

The future scope of sensor-based plant protection using Arduino is quite promising and can be explored in several ways:

1. Integration with Artificial Intelligence: Future systems could leverage AI algorithms to analyse data from sensors and make predictive recommendations for plant care. This could help in anticipating issues before they arise, such as pest infestations or nutrient deficiencies.
2. Advanced IoT Connectivity: As the Internet of Things (IoT) continues to evolve, these systems could become more interconnected. This would allow for better communication between devices, enabling smarter decision-making and automation across various agricultural practices.

3. Adaptation for Diverse Cultures: The technology can be adapted for different types of agriculture, including organic farming, vertical farming, and greenhouse management. This versatility can help meet the specific needs of various crops and growing environments.
4. Data Analytics and Machine Learning: The collection of large datasets over time can lead to insights that improve plant care techniques. Implementing machine learning could help in optimizing watering schedules, nutrient delivery, and other care practices based on historical data.
5. Sustainability and Resource Management: As environmental concerns grow, these systems can contribute to sustainable farming practices. By optimizing water and fertilizer use, they can help reduce waste and promote eco-friendly agriculture.

6. Mobile and Remote Applications: The development of mobile applications that allow farmers to monitor and control their systems remotely can enhance convenience and efficiency. This can also include alert systems for critical conditions, enabling timely interventions.
7. Community and Knowledge Sharing: Creating platforms for farmers to share data, experiences, and best practices can foster collaboration and innovation within agricultural communities, leading to collective improvements in crop management.

In summary, the future of sensor-based plant protection using Arduino holds great potential for advancing agricultural practices, enhancing productivity, and promoting sustainability.

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